



M-Tech Production & Marketing

Technology | Training | Marketing | Projects

TASK 1 REPORT

Internship Program — Batch 2026

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1. Introduction

This report presents the work completed for Experiment 03 of the M-Tech Production & Marketing AI/ML Internship Program, Batch 2026. The main focus of this task is to learn basic Data Handling and Analysis using two core Python libraries: NumPy and Pandas. M-Tech Production & Marketing, based in Haripur, KPK, provides a structured learning path, and this task builds the foundation for working with real-world datasets in Machine Learning.

2. Objectives

- Learn how to create and manage NumPy arrays, and understand their shapes and dimensions.
- Understand vectorized math operations to perform fast calculations without loops.
- Learn how to load, inspect, filter, and group tabular data using Pandas DataFrames.
- Handle missing data (NaN values) to clean datasets for Machine Learning models.
- Find basic statistics and learn data manipulation techniques.

3. Task Description

The task involves practicing data analysis basics. We need to work with multi-dimensional numbers using NumPy and manage structured tables using Pandas. The work includes writing Python scripts to perform array operations, filter data based on specific conditions, handle empty/missing blocks in tables, and group data by categories to find average values.

4. Work Done

I completed the lab session and ran all code exercises successfully. I practiced creating arrays, performing mathematical actions without loops, loading dynamic CSV files, and grouping data columns. All required five lab tasks were written and executed step-by-step in a Jupyter Notebook to verify their outputs.

5. Challenges Faced

- **Understanding Axes:** Understanding axis=0 (working down columns) vs axis=1 (working across rows) in NumPy was confusing at first.
- **Missing Data Errors:** Python throws errors or gives wrong results if data contains missing blocks (NaN). Learning how to find and fill them using .fillna() took some practice.
- **File Paths:** Loading the CSV file caused a small issue initially because the data file was not in the same folder as the Jupyter Notebook code.

6. Key Learnings

NumPy Speed: NumPy uses "vectorization", which means it processes entire arrays together. This is much faster than running a Python for loop on lists.

Pandas Utility: Pandas DataFrames work exactly like Excel sheets. They make it very easy to view, sort, filter, and clean data rows.

Data Prep: ML models cannot train on empty files or text data. Cleaning data by filling missing spots with the average (mean) value is a key step before building any model.

7. Conclusion

This lab successfully provided hands-on experience with data manipulation tools. NumPy and Pandas are highly efficient libraries that form the basis of all data science pipelines. By completing these tasks, I learned how to import real data files, filter essential information, fix missing data records, and summarize numbers effectively, making the dataset ready for machine learning algorithms.

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